Jahangirnagar University
জাহাঙ্গীরনগর বিশ্ববিদ্যালয়

Department of Computer Science and Engineering

Submission Date: 16/9/25

2nd Year 1st Semester (Hons)

Course Title: Digital Logic Design Laboratory

Course Code: CSE-254

Experiment No: ০৪

Experiment Title: Design and implementation of SR Latches, Gated Latches and D Latch.

Submitted To:

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16.9.25

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Experiment Name:

Design and implementation of S-R Latches (Active High and Active low), Gated S-R Latch and D-Latch.

Objectives:

To design, construct and verify the operation of:

- Active High S-R Latch (Using 2 NOR Gates)
- Active Low S-R Latch (Using NAND Gates)
- Gated S-R Latch
- D-Latch

Required components:

Digital trainer board, bread board, IC 7402, IC 7400, IC 7408, IC 7404, connecting wires.

Theory:

SR Latch: Basic bistable memory element

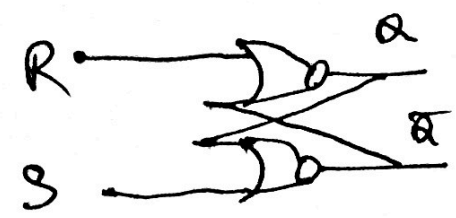
Active High: Inputs → S=1, resets Q=1, R=1; resets Q

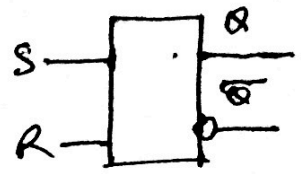
Active Low: Inputs → S=0, sets Q=1, R=0, resets Q.

Gated SR Latch: An enable input is added. (When E=1)

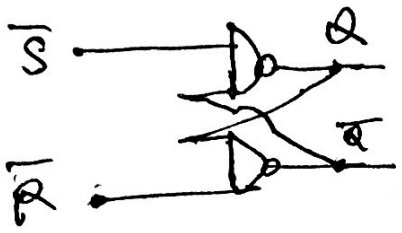
D-Latch: Built from an SR Latch.

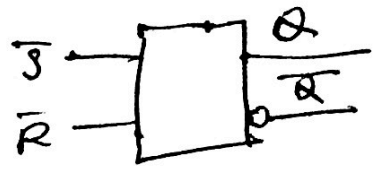
Block and Circuit Diagram:



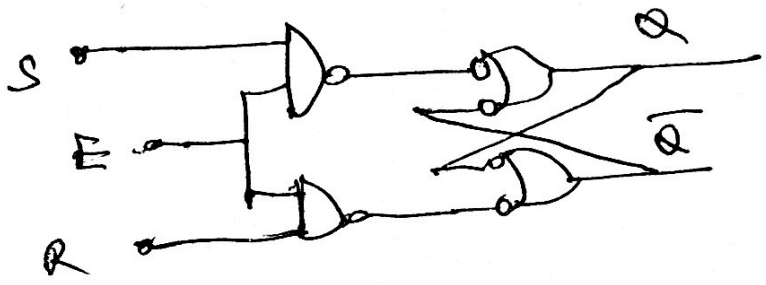


Active High S-R Latch

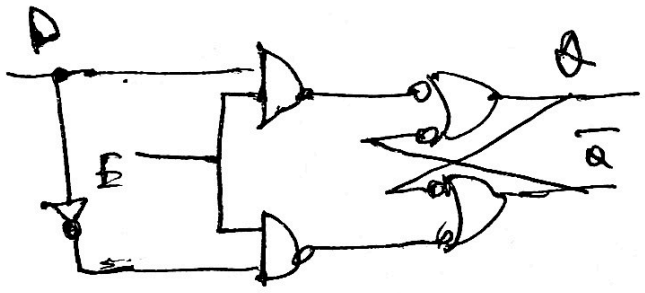




Active Low S-R Latch



Gated S R Latch



Truth Table:

S	R	Q^{t-1}	Q^t
0	0	0	NC
0	1	1	Reset
1	0	0	set
1	1	*	Inv

Active High S-R Latch

$\bar{S}$	$\bar{R}$	Q^{t-1}	Q^t
0	0	*	Inv
0	1	1	set
1	0	0	Reset
1	1	0	NC

Active Low S-R Latch

E	S	R	Q^{t-1}	Q^t
0	X	X	0	NC
1	0	0	1	NC
1	0	1	0	Reset
1	1	0	1	set
1	1	1	*	Inv

Gated SR Latch

E	D	Q^{t-1}	Q^t
0	X	1	NC
1	0	0	Reset
1	1	1	set

D-Latch

Procedure:

- Place ICs on the breadboard. Connect GND Pin 14 to Vcc and 7 to ground.
- Connect inputs (S, R, E, D) to switches.
- Connect outputs (Q, $\bar{Q}$) ~~via~~ to LED.
- verify truth tables.

Results:

- All latches functioned as per their truth table.
- Active High, Low responded correctly to SET/RESET
- Gated Latch enabled when (E = 1)
- D-Latch eliminated invalid state. Q followed when D=1, E=1.

Analysis:

- All latches stored 1 bit data correctly.
- Practical results matched theoretical truth table.
- D-Latch is more reliable than SR Latches.